

Curriculum Vitae

Jincheng Zhou

Personal Website: <https://www.jinchengzhou.net/>

Google Scholar: <https://scholar.google.com/citations?user=-3BQQJ4AAAAJ>
zhou791@purdue.edu, (213)706-6781

EDUCATION

Purdue University

West Lafayette, IN

Aug 22 – May 27 (expected)

- Ph.D. in Computer Science, **GPA: 3.93/4.00** (in progress)

- Advisor: Prof. Bruno Ribeiro

- Related Coursework: Computation & Learn on Graphs, Deep Learning, Theoretical CS Toolkit

University of Southern California

Los Angeles, CA

Aug 21 – July 22

- Master of Science in Computer Science, **GPA: 3.68/4.00**

- Related Coursework: Applied Natural Language Processing, Affective Computing, Foundations of Artificial Intelligence, Analysis of Algorithms

University of Southern California

Los Angeles, CA

Aug 17 – May 21

- Bachelor of Science in Computer Science, **GPA: 3.96/4.00**

- Graduated with *Computer Science Outstanding Student Award* (Top 10 students in CS Dept.)

- Related Coursework: Introduction to Artificial Intelligence, Deep learning and its Applications, Theory of Computing, Software Engineering

- Bachelor of Science in Mathematics, **GPA: 3.96/4.00**

Aug 17 – May 21

- Related Coursework: Mathematics of Machine Learning, Probability Theory, Mathematical Statistics, Analysis, Advanced Topics in Linear Algebra, Geometry and Transformations, Number Theory, Applied Combinatorics, Abstract Algebra, Numerical Methods

RESEARCH INTERESTS

- Causal Discovery, Causal Inference, Out-of-Distribution Generalization, Graph Representation Learning, Graph Foundation Models, LLM Agents, LLM Reasoning.

WORK EXPERIENCE

Intel Corporation

Graduate Technical Intern

Jan 2024 – Aug 2024

- Develop new methods to enhance Large Language Models (LLMs) reasoning and generalization performance on Knowledge Graphs (KGs) -informed tasks.
- Propose novel insights to tackle the challenges of how to properly integrate KGs and natural language inputs for LLMs in a principled manner.
- Investigate robust, permutation-equivariant Positional Encodings (PEs) for LLMs to interpret and understand symmetries inherent in knowledge graph structures.
- Demonstrate robust generalizability to real-world graphs with a variety of off-the-shelf open-sourced LLMs.
- Submit the results as a paper to a top-tier conference.

PUBLICATIONS (*: equal contribution)

- [1] Xinyu Luo*, **Jincheng Zhou***, Mengbo Wang, Zilin Shen, Nadia Atallah Lanman, Petros Drineas, Bruno Ribeiro, Brian Bullins. “Ori: Capturing Gradient Subspace Drift via Streaming k-PCA for Very Low-Rank LLM Training.” *In submission*.
- [2] **Jincheng Zhou**, Mengbo Wang, Anqi He, Yumeng Zhou, Hessam Olya, Murat Kocaoglu, Bruno Ribeiro. “Differentiable Constraint-Based Causal Discovery.” *NeurIPS 2025 (poster)*.
<https://neurips.cc/virtual/2025/poster/120335>
<https://arxiv.org/abs/2510.22031>
- [3] **Jincheng Zhou**, Yucheng Zhang, Mikhail Galkin, Bruno Ribeiro. “Robust Graph Positional Encodings for OOD Size Generalization in Transformers.” *In submission*.

- [4] Yangyi Shen, **Jincheng Zhou**, Beatrice Bevilacqua, Joshua Robinson, Charilaos Kanatsoulis, Jure Leskovec, Bruno Ribeiro, “Zero-Shot Generalization of GNNs over Distinct Attribute Domains.” *ICML 2025*. <https://openreview.net/forum?id=W4O4SwCjgY>
- [5] Mikhail Galkin, **Jincheng Zhou**, Bruno Ribeiro, Jian Tang, Zhaocheng Zhu. “Zero-shot Logical Query Reasoning on any Knowledge Graph.” *NeurIPS 2025 (poster)*. <https://neurips.cc/virtual/2024/poster/95712>
- [6] Bruna Jasinowodolinski, Yucheng Zhang, **Jincheng Zhou**, Beatrice Bevilacqua, Bruno Ribeiro. “Long-Range Synthetic Knowledge Graph Benchmarks for Double-Equivariant Models.” *ICLR 2024 BGPT Workshop*.
- [7] **Jincheng Zhou**, Beatrice Bevilacqua, and Bruno Ribeiro. “An OOD Multi-Task Perspective for Link Prediction with New Relation Types and Nodes.” *NeurIPS 2023 GLFrontiers Workshop*. <https://arxiv.org/abs/2307.06046>
- [8] **Jincheng Zhou***, Gao, Jianfei*, Yangze Zhou*, and Bruno Ribeiro. “Double Equivariance for Inductive Link Prediction for Both New Nodes and New Relation Types.” *NeurIPS 2023 GLFrontiers Workshop (Oral Presentation)*. <https://arxiv.org/abs/2302.01313>
- [9] **Jincheng Zhou**, and Volkan Ustun. “PySigma: Towards Enhanced Grand Unification for the Sigma Cognitive Architecture.” In *International Conference on Artificial General Intelligence*, pp. 355-366. Cham: Springer International Publishing, 2021. https://doi.org/10.1007/978-3-030-93758-4_36
- [10] Yunzhe Wang, Nikolos Gurney, **Jincheng Zhou**, David V. Pynadath, and Volkan Ustun. “Route Optimization in Service of a Search and Rescue Artificial Social Intelligence Agent.” In *AAAI Fall Symposium*, pp. 220-228. Cham: Springer Nature Switzerland, 2021. <https://drive.google.com/file/d/1SvstZZz8axwi-rcZfaqiHPXcHv1JlGmH/view?usp=sharing>

*equal contribution

TALKS & PRESENTATIONS

- *Double Equivariance for Inductive Link Prediction for Both New Nodes and New Relation Types*. Oral presentation at the NeurIPS 2024 GLFrontiers Workshop. Dec 15, 2023. URL: <https://nips.cc/virtual/2023/workshop/66500>
- *PySigma: Towards Enhanced Grand Unification for the Sigma Cognitive Architecture*. Technical presentation at the 14th Conference of Artificial General Intelligence. Oct 17, 2021. URL: <https://youtu.be/evXLHTFTgxQ>

SKILLS

- Programming languages: Python, C++, Julia, Java, JavaScript/TypeScript, HTML/CSS, SQL, SPARQL, Common Lisp, Haskell.
- Packages & Libraries: Slurm, PyTorch, PyTorch Geometrics (PyG), Deep Graph Library (DGL), TensorFlow, JAX, NumPy, Pandas, Scikit Learn.

FELLOWSHIP, AWARDS & HONORS

- **Ross Fellowship, Purdue University** April 2022
 - Guaranteed four (4) years of Ph.D. support valued at approximately \$214,778.00.
- **Computer Science Outstanding Student Award, USC Viterbi School of Engineering** May 2021
 - Top 10 students in the CS Department.
- **USC Viterbi Dean’s List** Dec 2019
- **USC Dornsife Dean’s List** Dec 2019, Dec 2018, May 2018
- **USC Academic Achievement Award** Dec 2019, Dec 2018, May 2018

RESEARCH EXPERIENCE & PROJECTS

Ph.D. Student at Purdue University

Jan 2023 – Sep 2023

An OOD Multi-Task Perspective for Link Prediction with New Relation Types and Nodes

Advisor: Prof. Bruno Ribeiro

- Identified a weakness in expressive power of the double equivariant graph representations as proposed in the paper “*Double Equivariance for Inductive Link Prediction for Both New Nodes and New Relation Types*,” by proving that said methods are unable to model certain conflicting patterns in knowledge graphs.

- Developed a theoretical framework, dubbed Multi-Task Double Exchangeability, that explains said conflicting patterns in knowledge graphs.
- Proposed a novel graph neural network architecture, named MTDEA, that is expressive enough model multi-task double exchangeability in knowledge graph.
- Proposed a novel test-time adaptation regime for adapting the MTDEA model on the observable portion of a given test graph to learn the conflicting patterns in the test graph.
- Conducted comprehensive experiments showing the superiority of the MTDEA model in modeling conflicting patterns in knowledge graphs compared to baselines.

Ph.D. Student at Purdue University

Jan 2023 – Sep 2023

Double Equivariance for Inductive Link Prediction for Both New Nodes and New Relation Types

Advisor: Prof. Bruno Ribeiro

- Solved a novel task of zero-shot doubly inductive link prediction in knowledge graphs (a.k.a. knowledge graph completion) with completely unknown test-time entities and relations and without any semantic or ontological information (e.g. textual descriptions) of the entities and relations.
- Developed a unifying theoretical framework, dubbed Double Equivariant Graph Representations that underlie any methods capable of solving this novel doubly inductive link prediction task.
- Proposed a novel graph neural network architecture, named ISDEA, that implements the double equivariant graph representations and is capable of accurately performing the doubly inductive link prediction task with robust performance.
- Proposed an easy-to-implement and theoretically sound improved version of an existing graph neural network architecture, named DEq-InGram, that significantly boosts the performance compared to the original InGram.
- Conducted comprehensive experiments showing the superiority of the ISDEA and DEq-InGram model in accurately performing the doubly inductive link prediction task compared to a wide range of baselines.

Student Worker/Research Assistant at the USC Institute for Creative Technologies

Jan 2020 – Aug 2022

PySigma: Towards Enhanced Grand Unification for the Sigma Cognitive Architecture

Supervisor/Mentor: **Prof. Paul Rosenbloom** and **Dr. Volkan Ustun**

- Redesigned the entire computational layer (the graphical layer) of the Sigma Cognitive Architecture.
- Conducted a comprehensive literature review on message-passing inference algorithms for probabilistic graphical models.
- Enabled probabilistic logical predicates to encode arbitrary probabilistic distributions and created new message structures to represent such predicates.
- Conceived a generalized message-passing algorithm combining *Variational Message Passing (VMP)* and *Monte Carlo Message Passing (MCMP)* for continuous approximate inference in lifted factor graphs.
- Analyzed the mathematical correctness of the said message design and generalized algorithm.
- Implemented the entire *PySigma* architecture from the ground up in Python using PyTorch while strictly following software engineering best practices such as comprehensive unit testing and documentation.

Student Worker/Research Assistant at the USC Institute for Creative Technologies

Jan 2020 – Aug 2022

Route Optimization in Service of a Search and Rescue Artificial Social Intelligence Agent

Supervisor/Mentor: **Prof. Paul Rosenbloom** and **Dr. Volkan Ustun**

- Investigated various *Graph Transformer* models for efficiently and approximately solving route optimization problems such as *Travelling Salesman Problem (TSP)* and *Capacitated Vehicle Routing Problem (CVRP)*.
- Designed a *Cooperative Multi-Agent Reinforcement Learning* system where agents with different roles and capabilities use the said *Graph Transformer* models as the oracle to get approximately optimal paths and cooperate to solve a Search-and-Rescue task in a Minecraft environment.
- Constructed an efficient data preprocessing pipeline for the Message-bus of the Search and Rescue Testbed.
- Streamlined the experimentation workflow by integrating the codebase with *Data Version Control (DVC)* for version-controlling experiments and *WandB* for real-time logging of the experiment statistics.

Undergrad Researcher at the USC Cognitive Learning for Vision and Robotics lab

Aug 2019 – Nov 2019

Generalizing Reinforcement Learning to Unseen Actions

Supervisor/Mentor: **Prof. Joseph Lim**

- Conceptualized and designed a variety of new tasks with vastly different properties for a novel RL benchmark environment.

- Conducted large-scale experiments studying the capability of various RL algorithms to generalize to unseen actions in the novel RL benchmark.
- Designed and built a dynamic webpage frontend for interactively showcasing the project.

Research Internship at the USC Institute for Creative Technologies

May 2019 – Aug 2019

Neural Learning of First-Order Probabilistic Logics in the Sigma Cognitive Architecture

Supervisor/Mentor: **Prof. Paul Rosenbloom** and **Dr. Volkan Ustun**

- Conducted a comprehensive literature review on neural approaches to learning first-order probabilistic logic.
- Enhanced the model proposed by *Neural Logic Machines (NLM)* in principled ways and adapted the model in the Sigma Cognitive Architecture.
- Conducted experiments at various scales and analyzed the pros and cons of the Sigma implementation of the *NLM* model.

Project Leader, Course Project for CSCI 599: Deep Learning and Its Applications

Jan 2019 – May 2019

Learn and Exploit Causality for Curiosity-Driven Deep Reinforcement Learning

Supervisor/Mentor: **Prof. Joseph Lim**

- Conducted a comprehensive literature review on curiosity / intrinsic motivation in reinforcement learning.
- Conceptualized and formalized the notion of causally inspired intrinsic motivation expressed by causal uncertainty.
- Developed a model-based reinforcement learning algorithm that learns a *Structured Causal Model (SCM)* of its environment and uses the model's causal uncertainty to guide the agent's exploration in the environment.

Undergrad Researcher at the USC Institute for Creative Technologies

Jan 2019 – May 2019

Deep Reinforcement Learning in the Sigma Cognitive Architecture

Supervisor/Mentor: **Prof. Paul Rosenbloom** and **Dr. Volkan Ustun**

- Compared and analyzed the performance of state-of-the-art RL algorithms under OpenAI Gym game environments.
- Implemented selected RL algorithms using Sigma Cognitive Architecture in Lisp and evaluated their performance.
- Extended Sigma's architectural capabilities in reinforcement learning, constructed infrastructure for building and benchmarking Sigma's RL implementations, and evaluated the architectural benefits of Sigma for reinforcement learning.

Undergrad Researcher at the USC Institute for Creative Technologies

Social Simulation of Human Behaviors

Sep 2018 – Dec 2018

Supervisor/Mentor: **Prof. David Pynadath**

- Analyzed pan-African socio-economic survey data from Afrobarometer
- Applied statistical analysis and data mining techniques to identify the correlations between people's actions and beliefs and uncover the underlying social dynamics within certain African communities
- Built *Partial Observable Markov Decision Process (POMDP)* agent models using *PsychSim* to simulate and reflect people's social behaviors and thought processes
- Devised a theoretical method to automate the construction of such a model on general public survey data

Other Selected Projects:

Project Leader, Computer Science Major Capstone Project

Feb 2021 – May 2021

LADA Open Discovery Toolkit LDD Genie

- Designed and implemented a full-stack open-sourced software to tag document boundaries and classify legal document types for the Los Angeles County District Attorney's Office
- Led the project team as the Person-of-Contact responsible for communicating with the stakeholders from the LADA Office and executing and maintaining an *Agile* software development pattern
- Built and tested a machine-learning model that automatically detects document boundaries and classifies document types, and integrated the model into the software to suggest solutions to user

TEACHING EXPERIENCE

- **Course Producer**(Undergrad TA) for USC CSCI 170: Discrete Methods in Computer Science
Aug 2018 – Dec 2018

- **Course Producer**(Undergrad TA) for USC CSCI 270: Introduction to Algorithms & Theory of Computing
Jan 2019 – May 2019
- **Course Producer**(Undergrad TA) for USC CSCI 270: Introduction to Algorithms & Theory of Computing
Aug 2019 – Dec 2019
- **Graduate Teaching Assistant** for Purdue CS58700: Foundations of Deep Learning.
Jan 2025 – May 2025